

An Exact Finite-Size Bulk–Edge and Quench Atlas for the SSH Chain

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Abstract

For the balanced open Su–Schrieffer–Heeger chain of M cells, we derive the full characteristic polynomial and prove that a strictly hyperbolic edge pair exists if and only if $w/v > (M+1)/M$. We give its exact splitting, two-ended eigenvectors, decay bound, critical linear taper, and fixed-ratio asymptotic. We then separate this finite threshold from the periodic bulk transition, resolve finite-ring parity and every hopping face, and derive an entire matrix-sinc propagator valid at singular couplings.

1 Finite hyperbolic edge threshold

Let $M \geq 2$ and $v, w \geq 0$. In the ordered basis $(A_1, \dots, A_M, B_1, \dots, B_M)$, the open Hamiltonian is

$$H_O = \begin{pmatrix} 0 & T \\ T^* & 0 \end{pmatrix}, \quad T = vI + wS_-, \quad (S_-)_{j,j-1} = 1. \quad (1)$$

Thus $\Gamma = \text{diag}(I, -I)$ obeys $\Gamma H_O \Gamma = -H_O$.

Theorem 1 (Characteristic and sharp finite edge theorem). *Put $q_M(y) = \det(yI - TT^*)$. Then*

$$q_0 = 1, \quad q_1 = y - v^2, \quad q_m = (y - v^2 - w^2)q_{m-1} - v^2w^2q_{m-2}. \quad (2)$$

For $v, w > 0$, set $x = (E^2 - v^2 - w^2)/(2vw)$. With U_m the Chebyshev polynomial of the second kind,

$$\det(EI - H_O) = (vw)^M \left[U_M(x) + \frac{w}{v} U_{M-1}(x) \right]. \quad (3)$$

There is exactly one secular root $x = -\cosh \kappa < -1$, and hence exactly one eigenvalue pair $\pm E_e$ outside the trigonometric sector, if and only if

$$\frac{w}{v} > \frac{M+1}{M}. \quad (4)$$

For the unique $\kappa > 0$ determined by

$$\frac{w}{v} = \frac{\sinh((M+1)\kappa)}{\sinh(M\kappa)}, \quad (5)$$

the positive energy and one unnormalised positive-energy vector are

$$E_e = v \frac{\sinh \kappa}{\sinh(M\kappa)} = w \frac{\sinh \kappa}{\sinh((M+1)\kappa)}, \quad (6)$$

$$a_j = (-1)^{j-1} \sinh((M+1-j)\kappa), \quad b_j = (-1)^{j-1} \sinh(j\kappa). \quad (7)$$

Thus (a, b) has energy E_e and $(a, -b)$ has energy $-E_e$. The profiles decay strictly inward:

$$\frac{|a_{j+1}|}{|a_j|} < e^{-\kappa}, \quad \frac{|b_j|}{|b_{j+1}|} < e^{-\kappa} \quad (1 \leq j < M). \quad (8)$$

At equality in (4), $x = -1$ and $E = v/M$. After common division of a, b by κ (equivalently, after overall normalisation), the $\kappa \downarrow 0$ profiles are $a_j = (-1)^{j-1}(M+1-j)$ and $b_j = (-1)^{j-1}j$.

Proof. The leading diagonal entry of TT^* is v^2 ; every subsequent one is $v^2 + w^2$, and the off-diagonal entries are vw . Its continuant is (2). Normalising this recurrence gives (3); in particular, the factor $(vw)^M$ cannot be dropped. The M roots are real and simple because TT^* is an irreducible Jacobi matrix. Also $\det T = v^M$, so $E = 0$ is impossible when $v > 0$.

For $x = -\cosh \kappa$, the identity $U_m(-\cosh \kappa) = (-1)^m \sinh((m+1)\kappa)/\sinh \kappa$ reduces the secular equation to (5). Its logarithmic derivative is

$$(M+1) \coth((M+1)\kappa) - M \coth(M\kappa) > 0,$$

because $s \mapsto s \coth(s\kappa)$ is strictly increasing. The ratio therefore rises from $(M+1)/M$ to infinity. No root lies above $x = 1$, where both Chebyshev terms are positive, so this proves the iff statement and uniqueness. Direct hyperbolic addition verifies $Tb = E_e a$ and $T^*a = E_e b$, including both expressions in (6). Finally,

$$\frac{\sinh(n\kappa)}{\sinh((n+1)\kappa)} = e^{-\kappa} \frac{1 - e^{-2n\kappa}}{1 - e^{-2(n+1)\kappa}} < e^{-\kappa},$$

which proves (8). Taking $\kappa \downarrow 0$ gives the threshold energy; applying the common $1/\kappa$ rescaling before that limit gives the nonzero tapers. $\square$

Corollary 2 (Finite splitting). *For fixed $r = w/v > 1$ and all sufficiently large M , the hyperbolic root exists, $\kappa_M \rightarrow \log r$, and*

$$E_e \sim w(1 - r^{-2})r^{-M}. \quad (9)$$

Consequently the finite pair is never exactly zero for $v > 0$, although its splitting is exponentially small.

Proof. Writing $z = e^{-\kappa}$ turns (5) into $r = z^{-1}(1 - z^{2M+2})/(1 - z^{2M})$, hence $z \rightarrow r^{-1}$. Equation (6) becomes $E_e = vz^{M-1}(1 - z^2)/(1 - z^{2M})$, which gives (9). $\square$

2 Bulk winding, boundary faces, and unitary propagation

Close the chain and use momenta $k_n = 2\pi n/M$. Our orientation convention is

$$q(k) = v + we^{ik}, \quad h(k) = \begin{pmatrix} 0 & \overline{q(k)} \\ q(k) & 0 \end{pmatrix}. \quad (10)$$

Theorem 3 (Periodic bulk and finite parity). *The Bloch energies are*

$$E_{\pm}(k) = \pm \sqrt{v^2 + w^2 + 2vw \cos k}. \quad (11)$$

For the continuum symbol and the sampled M -cell ring, respectively, let Δ_{∞} and Δ_M denote the minimum positive-band distance to zero. Then

$$\Delta_{\infty} = |v - w|, \quad \Delta_M = \begin{cases} |v - w|, & M \text{ even}, \\ \sqrt{v^2 + w^2 - 2vw \cos(\pi/M)}, & M \text{ odd}. \end{cases} \quad (12)$$

The corresponding central band gaps are $2\Delta_{\infty}$ and $2\Delta_M$. Whenever the loop avoids zero, its counterclockwise winding is 1 for $w > v$ and 0 for $v > w$; at $v = w > 0$ it is undefined. At this critical line a finite ring has a two-dimensional zero sector if and only if M is even. For odd M it instead has $\Delta_M = 2v \sin(\pi/(2M)) > 0$.

Proof. Squaring (10) gives $|q(k)|^2 I$, proving the dispersion. Over the continuum, $\cos k$ is minimized at $k = \pi$. The finite grid samples that point exactly for even M ; for odd M its nearest points are $\pi \pm \pi/M$, which proves (12). The curve $q(k)$ is the counterclockwise circle of radius w centred at v . It encloses zero exactly when $w > v$. On the critical line it reaches zero only at $k = \pi$, which belongs to the finite grid exactly for even M ; the corresponding 2×2 fiber is zero. $\square$

Theorems 1 and 3 exhibit a finite separation: the bulk is topological for $w/v > 1$, but the open pair is strictly hyperbolic only for $w/v > (M+1)/M$. Equality in the latter condition is a band-edge taper, not an exponentially localized state.

Theorem 4 (All hopping faces). *For the open chain the following statements are exact.*

1. *If $w = 0 < v$, there are M dimers and $\pm v$ each has multiplicity M .*

2. If $v = 0 < w$, there are $M - 1$ dimers, $\pm w$ each has multiplicity $M - 1$, and A_1, B_M span a two-dimensional zero sector.
3. If $v = w = 0$, the kernel has dimension $2M$.
4. If $v = w > 0$, then $\text{spec}(H_O) = \{2v \cos(\ell\pi/(2M + 1)) : 1 \leq \ell \leq 2M\}$, so zero is absent.

For $M = 1$, declared separately, the open intercell bond is absent and the energies are $\pm v$; the periodic wrap merges with the intracell bond and the energies are $\pm(v + w)$.

Proof. The first three statements follow by listing the isolated sites and bonds. The fourth is the sine diagonalisation of the uniform path on $2M$ sites; its denominator $2M + 1$ prevents a zero cosine. The last statement follows from the two explicit 2×2 matrices. $\square$

Theorem 5 (Entire boundary-safe propagator). *Let $\text{sinc } z = \sin(z)/z$ with $\text{sinc } 0 = 1$. Entire matrix functional calculus gives, on every face in Theorem 4,*

$$e^{-itH_O} = \begin{pmatrix} \cos(t\sqrt{TT^*}) & -it \text{sinc}(t\sqrt{TT^*})T \\ -it \text{sinc}(t\sqrt{T^*T})T^* & \cos(t\sqrt{T^*T}) \end{pmatrix}. \quad (13)$$

It is unitary and satisfies $\Gamma e^{-itH_O} \Gamma = e^{itH_O} = (e^{-itH_O})^$.*

Proof. The even and odd powers have block forms

$$H_O^{2n} = \text{diag}((TT^*)^n, (T^*T)^n), \quad H_O^{2n+1} = \begin{pmatrix} 0 & (TT^*)^n T \\ (T^*T)^n T^* & 0 \end{pmatrix}.$$

Separating the exponential series proves (13). Because cosine and sinc are entire in the squared operators, no inverse or positive singular-value assumption enters. Unitarity follows from self-adjointness, and the last identity from chirality. $\square$

Source lineage

References

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